Quasi-Monte Carlo for the European Call: Halton, Sobol, and RQMC

Quant Finance Portfolio — Phase 2, Block 3.1

May 6, 2026

Abstract

Building on the foundations established in Block 3.0, we apply quasi-Monte Carlo to the European call pricer. To make QMC non-trivially multi-dimensional we use the multi-step Euler discretisation of Block 1.2.1 with $N = 20$ time steps, giving an integration problem of dimension $d = 20$. We compare four estimators: IID Monte Carlo (baseline from Block 1.2.1), QMC with Halton, QMC with Sobol, and randomized QMC (RQMC) with Sobol plus digital shifting. The empirical convergence rates measured on a log-log slope plot are expected to confirm the theoretical predictions: slope ≈ -0.5 for IID, slope close to -1 for Sobol RQMC at this dimension. The validation script also confirms that RQMC delivers a usable half-width via R-replication averaging.

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1 The integration problem

The Euler-discretised European call price is

$$\mathbb{E}^{\mathbb{Q}}\big[e^{-rT}\max(S_T^{(N)} - K, 0)\big], \tag{1}$$

where $S_T^{(N)} = S_0 \prod_{k=1}^{N}(1 + rh + \sigma\sqrt{h}\, Z_k)$ with $h = T/N$ and $Z_1, \ldots, Z_N$ i.i.d. standard normals. With $N = 20$ this is a 20-dimensional integral against a Gaussian measure. Equivalently, it is a 20-dimensional integral against the uniform measure on $[0, 1)^{20}$ via the change of variables $Z_k = \Phi^{-1}(u_k)$:

$$C^{\text{Euler}} \;=\; \int_{[0,1)^{20}} e^{-rT}\max\big(S_T^{(N)}(\Phi^{-1}(u_1), \ldots, \Phi^{-1}(u_{20})) - K,\; 0\big)\,\mathrm{d}u. \tag{2}$$

This is the integral that QMC will attack.

Why $N = 20$ rather than $N = 1$? The exact GBM sampler of Block 1.1 needs only one normal per path, so the European call under exact GBM is a 1-dimensional integral. QMC at $d = 1$ is trivial (van der Corput) and not pedagogically rich. Choosing the Euler discretisation of Block 1.2.1 with $N = 20$ gives a genuine 20-dimensional problem where QMC's advantage matters.

The difference between exact and Euler is the $O(h)$ weak bias documented in Block 1.2.1; for $N = 20$ this bias is approximately 0.05 in absolute terms, large enough to be visible against MC error at $n \sim 10^4$ but small enough not to obscure the convergence-rate analysis. We accept this bias as a fixed offset and study the rate at which the Monte Carlo / QMC error around C^{Euler} vanishes as n grows.

2 The four estimators

We construct four estimators of C^{Euler}, all sharing the same Φ^{-1} inversion step but differing in how the underlying $u_i \in [0, 1)^{20}$ points are produced.

IID-MC. Standard Monte Carlo. The points u_i are pseudo-random uniform draws from a Mersenne-Twister generator. This is the Block 1.2.1 baseline.

QMC-Halton. The points u_i are the first n entries of the 20-dimensional Halton sequence with primes $p_1 = 2, p_2 = 3, \ldots, p_{20} = 71$. The estimator returns a single number without a half-width.

QMC-Sobol. The points u_i are the first n entries of the 20-dimensional Sobol sequence with Joe-Kuo direction numbers. As with Halton, the deterministic estimator has no half-width.

RQMC-Sobol. The points are the Sobol sequence randomised by digital shifting with R independent shifts (we use $R = 20$). Each replication produces an estimate of dimension $d = 20$ from n Sobol points; the R replication estimates are i.i.d. and a half-width is constructed from their sample variance.

2.1 The fixed and varying axes

To compare convergence rates, we vary the per-replication path count n across a logarithmic grid. The rest of the configuration is fixed:

- Euler steps: $N = 20$.
- Total dimensionality: $d = 20$.
- RQMC replications: $R = 20$.
- Random seed for IID and for RQMC's shifts: fixed for reproducibility.
- Target: the same canonical parameters $S_0 = K = 100$, $r = 0.05$, $\sigma = 0.20$, $T = 1$.

The reference value against which we measure error is the closed-form Black-Scholes price $C^{\text{BS}} = 10.4506$. The Euler scheme's $O(h)$ bias creates a systematic offset of approximately 0.05 from C^{BS}; this offset is the *same for all four estimators* (they all share the same discretised payoff structure), so when we plot "error vs n" on a log-log scale, the bias contributes a horizontal asymptote that all four lines share. The slopes at moderate n, before the asymptote dominates, are what we are after.

3 Convergence rates: predictions

The theoretical asymptotic rates from Block 3.0:

- IID-MC: error $= O(n^{-1/2})$. Log-log slope -0.5.
- QMC-Halton: error $= O((\log n)^{20}/n)$. The $(\log n)^{20}$ factor at $n = 10^4$ is $(9.2)^{20} \approx 10^{19}$, so the constant in the big-O is huge and Halton may not yet be in its asymptotic regime at the budgets we test. Empirically, we expect a slope between -0.7 and -0.9 rather than the asymptotic -1.
- QMC-Sobol (deterministic): same asymptotic rate as Halton, but with substantially better constants thanks to the $\mathbb{F}_2$ construction. Empirically, we expect a slope close to -1 at moderate n.
- RQMC-Sobol: same expected slope as deterministic Sobol; the randomisation provides the half-width without affecting the rate.

3.1 The dimension-effect caveat

A naive reading of $(\log n)^d/n$ at $d = 20$ would predict QMC to require astronomical n to outperform MC. The empirical behaviour is much friendlier, for two reasons:

Effective dimension. The European call payoff, expressed in terms of $u \in [0,1)^{20}$, depends primarily on a small number of "effective dimensions": the few coordinates whose variation drives most of the variance of $S_T^{(N)}$. For the Euler discretisation, all 20 steps contribute roughly equally to $\log S_T^{(N)}$, so the effective dimension is closer to 20 than to 1, but the integrand's smoothness ($\max(\cdot, 0)$ has a single kink in $\mathbb{R}^{20}$) helps QMC's convergence in practice. The "intrinsic" difficulty is lower than the nominal $d = 20$ would suggest.

Brownian bridge construction. A standard QMC optimisation in finance is to construct the path of W_t in the *Brownian bridge* order rather than the natural left-to-right order: assign the first dimension of u to W_T (the most "important" coordinate), the second to $W_{T/2}$, the third and fourth to $W_{T/4}$ and $W_{3T/4}$, and so on. This concentrates the "effective dimension" in the first few QMC coordinates, where QMC's discrepancy advantage is strongest. *We do not apply the Brownian bridge construction in this writeup*: the natural left-to-right Euler discretisation already gives observable QMC advantage and the bridge optimisation can be added in a future block (it is most relevant for path-dependent options where the effective dimension can be much larger).

3.2 The convergence rate test

The validation script computes the absolute error

$$\mathrm{err}(n) \;=\; \left|\hat{C}_n - C^{\mathrm{Euler}}\right| \tag{3}$$

on a logarithmic grid $n \in \{10^3, 10^{3.5}, 10^4, 10^{4.5}, 10^5\}$ (five points), fits a linear regression of $\log \mathrm{err}$ on $\log n$, and reports the slope. We use the bias-corrected target C^{Euler} (estimated from a high-precision IID-MC run) rather than C^{BS} to avoid contaminating the rate measurement with the Euler bias.

4 Implementation overview

The Python implementation introduces a new module `quantlib.qmc`:

- `halton(n, d)`: returns the first n entries of the d-dimensional Halton sequence as an (n, d) NumPy array. Implemented from scratch using the radical inverse function.
- `sobol(n, d, scramble=False, seed=None)`: returns the first n Sobol points. Wraps `scipy.stats.qmc.Sobol`, which uses Joe-Kuo direction numbers internally.
- `mc_european_call_euler_qmc`: deterministic QMC pricer that returns a single estimate (no half-width). Accepts a parameter `sequence` $\in$ {`'halton'`, `'sobol'`}.
- `mc_european_call_euler_rqmc`: randomized QMC pricer with R digital shifts. Returns a full `MCResult` with the half-width estimated from the R replications.

The pricers reuse the Euler step from `quantlib.gbm`: only the source of normals changes (from `rng.standard_normal(...)` to Φ^{-1} applied to a QMC sequence). This is exactly the modular separation foreseen in Block 0 when we adopted inversion-based normal sampling.

5 Empirical validation strategy

The validation script `python/validate_mc_european_qmc.py` runs four tests.

Test 1: convergence rate slopes. Compute the absolute error of each estimator (IID, Halton, Sobol-det, Sobol-RQMC) at $n \in \{10^3, 10^{3.5}, 10^4, 10^{4.5}, 10^5\}$. Linear-regress $\log \mathrm{err}$ on $\log n$. Predicted slopes:

- IID: ≈ -0.5.
- Halton: ≈ -0.7 to -0.9.
- Sobol-det: ≈ -1.0 (or slightly less).
- Sobol-RQMC: ≈ -1.0 (same as deterministic Sobol; the randomisation only adds a small constant overhead).

Test 2: BS-coherence of RQMC-Sobol. The RQMC pricer at $n = 10^4$ paths and $R = 20$ replications agrees with C^{BS} modulo the Euler bias, within a few half-widths.

Test 3: half-width below MC at equal budget. RQMC-Sobol at $n_{\mathrm{paths}} \times R = 100\,000$ payoffs has half-width significantly below the IID-MC half-width at the same total budget.

Test 4: input validation. Mechanical: the QMC and RQMC pricers reject bad inputs.

6 Bibliographic notes

The application of QMC to financial option pricing is reviewed in Glasserman [2, Chapter 5]. The Brownian bridge construction in QMC is due to Caflisch, Morokoff and Owen [1]; we do not implement it here. The analysis of effective dimension in finance is in Wang and Sloan [3].

References

[1] R. E. Caflisch, W. Morokoff, A. B. Owen, *Valuation of mortgage-backed securities using Brownian bridges to reduce effective dimension*, Journal of Computational Finance, 1(1):27–46, 1997.

[2] P. Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer, 2004.

[3] X. Wang and I. H. Sloan, *Why are high-dimensional finance problems often of low effective dimension?*, SIAM Journal on Scientific Computing, 27(1):159–183, 2005.